

Archery Performance Analysis

The Relationship Between Experience and Shooting Accuracy

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Contents

1	Executive Summary	3
2	Introduction	3
2.1	Background	3
2.2	Analysis Approach	3
3	Data Preparation	4
4	Setup	4
4.1	Loading and Inspecting the Data	4
4.2	De-identification	5
4.3	Calculating Archery Experience	5
4.4	Separating Tournament Results	5
4.5	Variable Types	6
4.6	Data Cleaning and Taming	6
4.7	Reproducible Sampling	8
4.8	Calculating Accuracy	9
4.9	Descriptive Statistics	10
4.10	Standardising Numerical Variables	10
4.11	Comparing Standardised Variables	11
5	Variable with higher mean.	11
6	Variable with higher median.	11

7	Variable with higher inter-quartile range	11
8	Plot histograms of each variable and calculate the skewness.	12
8.1	Experience and Accuracy Relationship	13
8.2	Box-Cox Transformation	14
8.3	Applying the Transformation	15
8.4	Transformed Data	15
8.5	Population Linear Model	17
8.6	Sample Regression Line	18
8.7	Fitting the Linear Model	18
9	Predictions experience of 2, 5, 10, 15, 20 and 25 years.	21
10	Discussion and Conclusion	22
10.1	Interpretation	22
10.2	Limitations	23
10.3	Conclusion	23
10.4	Appendix: Reproducible R Analysis	24

1 Executive Summary

This report investigates the relationship between **archery experience and shooting accuracy** using tournament data from an archery club. The analysis considers how accuracy changes with the number of days an archer has practised and uses a linear model to make predictions at selected experience levels.

The original dataset contained 555 archers. After de-identification and data cleaning, two observations with unrealistic experience values were removed, leaving 553 observations. A reproducible random sample of 450 archers was then selected for the analysis.

Accuracy was calculated as the proportion of arrows that hit the target. Exploratory analysis showed a strong positive relationship between experience and accuracy, although the relationship appeared to become less steep at higher levels of experience. A Box-Cox transformation with an estimated $\lambda = 2.4$ was applied before fitting the linear model.

The fitted model achieved an R^2 of approximately **0.947**. Predicted accuracy increased from **42.4% after 2 years** of experience to **88.5% after 25 years**. These results suggest that greater experience is strongly associated with higher accuracy in this sample. However, the analysis does not establish a fixed number of years required to become a “good archer”, because the definition of good performance is subjective and other factors affecting performance are not included in the model.

Key takeaway: Experience is strongly associated with archery accuracy in this dataset, but the model should be interpreted only within a realistic range of experience.

2 Introduction

2.1 Background

The archery club would like to provide potential recruits with a quantitative answer to the question: **“How long does it take to become a good archer?”**

This question is not directly well-posed for statistical analysis because “good archer” does not have a defined performance threshold. Therefore, this analysis uses **shooting accuracy**, calculated as the proportion of arrows that hit the target, as a measurable indicator of performance.

2.2 Analysis Approach

The analysis follows the required Data Taming workflow: loading and inspecting the data, de-identifying participants, transforming and cleaning the variables, drawing a reproducible random sample, exploring the distributions, examining the relationship between experience and accuracy, fitting a linear model, checking its assumptions, and making predictions with 90% confidence intervals.

3 Data Preparation

The following sections document the data preparation and exploratory steps required for the analysis. The original R code has been retained so that the analysis remains fully reproducible.

4 Setup

```
library(tidyverse)
library(inspectdf)
library(stringr)
library(lubridate)
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.3
```

```
library(moments)
```

```
## Warning: package 'moments' was built under R version 4.5.2
```

```
library(tinytex)
```

4.1 Loading and Inspecting the Data

```
archery_raw <- read_csv("./data/archery_2.csv")
archery_raw
```

```
## # A tibble: 555 x 3
##   Archer   Commenced RES
##   <chr>   <chr>      <chr>
## 1 Skye    13/04/2008 Target hit 45 times from 55 shots.
## 2 Madison 25/09/2019 Target hit 29 times from 55 shots.
## 3 Chelsea 20/12/2006 Target hit 39 times from 52 shots.
## 4 William 15/02/2004 Target hit 49 times from 60 shots.
## 5 Lennox  15/07/2021 Target hit 25 times from 60 shots.
## 6 Brendan 9/05/2014  Target hit 39 times from 60 shots.
## 7 Jackson 29/04/2005 Target hit 41 times from 52 shots.
## 8 Asher   25/10/1996 Target hit 51 times from 54 shots.
## 9 Nicole  4/01/2007  Target hit 56 times from 69 shots.
## 10 Dylan  10/04/2014 Target hit 38 times from 59 shots.
## # i 545 more rows
```

```
dim(archery_raw) #dimention of the data set
```

```
## [1] 555    3
```

4.2 De-identification

```
archery_clean <- archery_raw  
archery_clean <- mutate(archery_clean, "ID"=1:555) #add unique id column  
archery_clean <- relocate(archery_clean,"ID",.before=Archer)#relocate id to far left  
archery_clean <-mutate(archery_clean,Archer=NULL)# remove Archery column
```

Remove column Archer to remove the name of the archers.

4.3 Calculating Archery Experience

```
archery_clean <- mutate(archery_clean,"XP"=dmy("16/06/2024")-dmy(archery_clean$Commenced))  
archery_clean <- relocate(archery_clean,"XP",.before=RES) #relocate clolumn XP
```

I included the day of tournament as there was a practice session before tournament. It should be considered as 1 day of experience before the tournament. I shifted the date from 15 June to 16 June to include the the tournament day as experience.

4.4 Separating Tournament Results

```
archery_clean <- mutate(archery_clean,"Arrows"=str_match(RES,"from (\\d+) shots.")[,2],  
                        "Targets"=str_match(RES,"hit (\\d+) times")[,2])  
# add two columns Arrows and Target  
archery_clean <- mutate(archery_clean,RES=NULL) #delete column residual.  
archery_clean
```

```
## # A tibble: 555 x 5  
##       ID Commenced  XP      Arrows Targets  
##   <int> <chr>      <drtn>    <chr>  <chr>  
## 1     1 13/04/2008 5908 days 55     45  
## 2     2 25/09/2019 1726 days 55     29  
## 3     3 20/12/2006 6388 days 52     39  
## 4     4 15/02/2004 7427 days 60     49  
## 5     5 15/07/2021 1067 days 60     25  
## 6     6  9/05/2014 3691 days 60     39  
## 7     7 29/04/2005 6988 days 52     41  
## 8     8 25/10/1996 10096 days 54     51
```

```
## 9      9 4/01/2007    6373 days 69      56
## 10     10 10/04/2014   3720 days 59      38
## # i 545 more rows
```

```
dim(archery_clean)
```

```
## [1] 555    5
```

4.5 Variable Types

- ID is categorical ordinal. It is assigned each archery to keep track of change in their data. Hence, the order is important.
- Commenced is categorical ordinal. It indicates the starting date of practice for each archery. Therefore, order is important to let us know the duration of experience each archery has.
- Arrows is quantitative discrete. It is a whole number and represents the total number of arrows an archery shot.
- Targets is quantitative discrete. It is also a whole number that represent the number of target hit.

4.6 Data Cleaning and Taming

```
archery_clean <- rename(archery_clean,id=ID)
archery_clean <- rename(archery_clean,commenced=Commenced)
archery_clean <- rename(archery_clean,xp=XP)
archery_clean <- rename(archery_clean,arrows=Arrows)
archery_clean <- rename(archery_clean,targets=Targets)
archery_clean$xp <- as.integer(archery_clean$xp)
archery_clean$arrows <- as.integer(archery_clean$arrows)
archery_clean$targets <- as.integer(archery_clean$targets)
archery_clean$id <- as.ordered(archery_clean$id)
archery_clean$commenced <- dmy(archery_clean$commenced) # change to date
inspect_na(archery_clean) #check if there is na data
```

```
## # A tibble: 5 x 3
##   col_name    cnt  pcnt
##   <chr>      <int> <dbl>
## 1 id          0      0
## 2 commenced   0      0
## 3 xp          0      0
## 4 arrows      0      0
## 5 targets     0      0
```

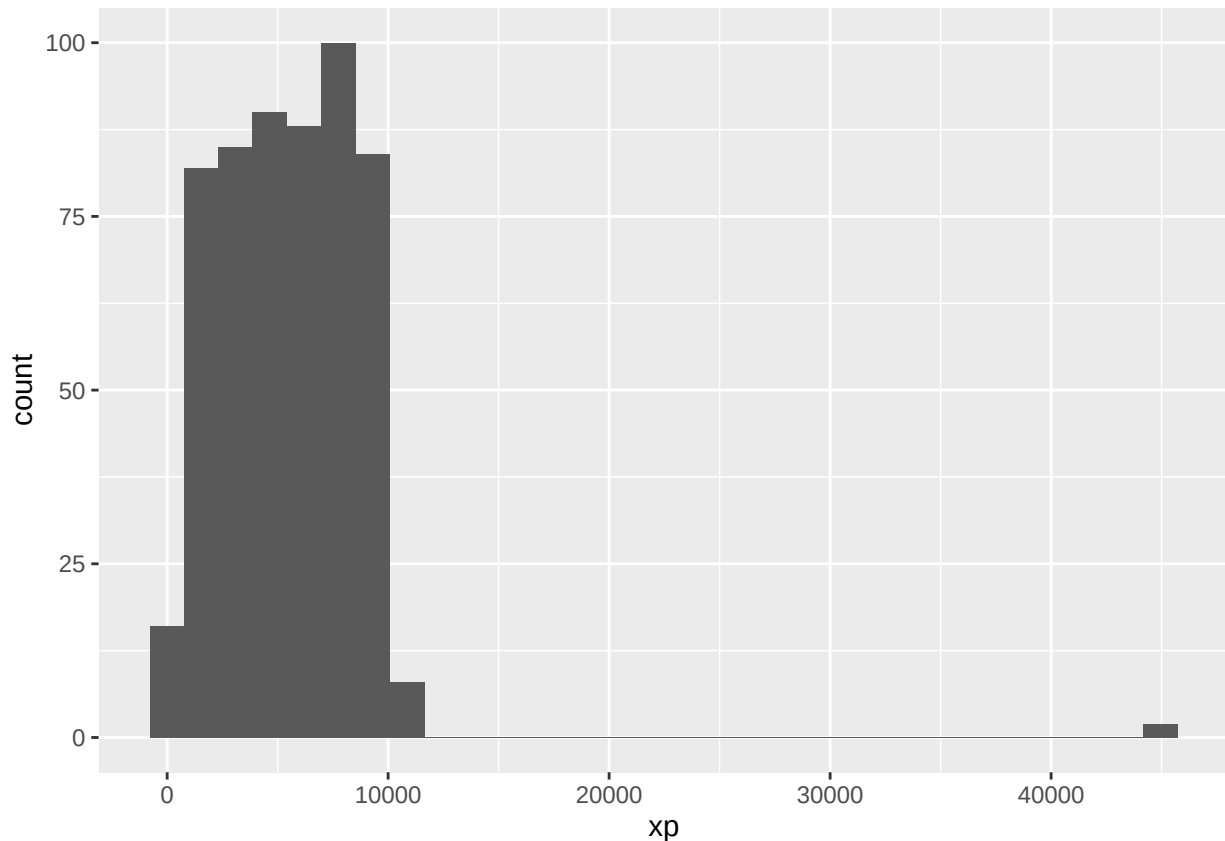
```
filter(archery_clean,(arrows>70|arrows<0))# check for out of range data for arrows
```

```
## # A tibble: 0 x 5
## # i 5 variables: id <ord>, commenced <date>, xp <int>, arrows <int>,
## #   targets <int>
```

```
filter(archery_clean,(targets>70|targets<0))# check for out of range data for targets
```

```
## # A tibble: 0 x 5
## # i 5 variables: id <ord>, commenced <date>, xp <int>, arrows <int>,
## #   targets <int>
```

```
ggplot(archery_clean, aes(xp))+ #check for abnormal data
  geom_histogram()
```



```
abn_data <- filter(archery_clean,xp>20000)
abn_data
```

```
## # A tibble: 2 x 5
##   id      commenced      xp arrows targets
```

```
##   <ord> <date>      <int> <int>   <int>
## 1 167   1900-01-01 45457    59     43
## 2 275   1900-01-01 45457    69     39
```

```
archery_clean <- filter(archery_clean, xp < 20000) # remove unrealistic xp
archery_clean
```

```
## # A tibble: 553 x 5
##   id      commenced      xp arrows targets
##   <ord> <date>      <int> <int>   <int>
## 1 1      2008-04-13  5908    55     45
## 2 2      2019-09-25  1726    55     29
## 3 3      2006-12-20  6388    52     39
## 4 4      2004-02-15  7427    60     49
## 5 5      2021-07-15  1067    60     25
## 6 6      2014-05-09  3691    60     39
## 7 7      2005-04-29  6988    52     41
## 8 8      1996-10-25 10096    54     51
## 9 9      2007-01-04  6373    69     56
## 10 10     2014-04-10  3720    59     38
## # i 543 more rows
```

```
dim(archery_clean)
```

```
## [1] 553    5
```

Rename the variables to lower case according to The Tame Data Conventions.

- id is categorical ordinal variable so it should be <ord>
- commenced: change to The Tame data Conventions format year-month-day and <date>
- xp, arrows and targets: change to <int>

Inspect NA to check if there is any data missing. Filter both targets and arrows to find if there is any data out of range. arrows and targets cannot be negative number and more than 70 as each archer is given up to 70 arrows to shoot.

id 167 and 275 have 45457 days experience which is equivalent to 125 years of experience. This is unrealistic. Hence, we remove the observations.

4.7 Reproducible Sampling

```
set.seed(1896853)
archery <- sample_n(archery_clean, 450) # a random sample of 450 archers
archery
```

```
## # A tibble: 450 x 5
##   id      commenced      xp arrows targets
##   <ord> <date>      <int> <int>   <int>
## 1 2      2019-09-25  1726    55     29
## 2 140    2019-01-16  1978    60     33
## 3 185    1997-01-14 10015    65     61
## 4 469    2012-11-25  4221    57     37
## 5 153    2017-08-08  2504    61     36
## 6 172    2003-08-12  7614    62     53
## 7 412    2004-12-31  7107    57     48
## 8 517    1997-03-09  9961    62     56
## 9 424    2003-04-19  7729    66     58
## 10 184   2018-09-20  2096    55     30
## # i 440 more rows
```

```
dim(archery)
```

```
## [1] 450 5
```

4.8 Calculating Accuracy

```
archery <- mutate(archery, "acc"=targets/arrows)
archery
```

```
## # A tibble: 450 x 6
##   id      commenced      xp arrows targets  acc
##   <ord> <date>      <int> <int>   <int> <dbl>
## 1 2      2019-09-25  1726    55     29 0.527
## 2 140    2019-01-16  1978    60     33 0.55
## 3 185    1997-01-14 10015    65     61 0.938
## 4 469    2012-11-25  4221    57     37 0.649
## 5 153    2017-08-08  2504    61     36 0.590
## 6 172    2003-08-12  7614    62     53 0.855
## 7 412    2004-12-31  7107    57     48 0.842
## 8 517    1997-03-09  9961    62     56 0.903
## 9 424    2003-04-19  7729    66     58 0.879
## 10 184   2018-09-20  2096    55     30 0.545
## # i 440 more rows
```

```
dim(archery)
```

```
## [1] 450 6
```

acc is quantitative continuous. It is derived from the ratio of two integers. The data type in the tibble is correct. There is no need to change.

4.9 Descriptive Statistics

```
inspect_num(archery)
```

```
## # A tibble: 4 x 10
##   col_name  min      q1  median   mean     q3    max      sd pcnt_na
##   <chr>    <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 xp      490   2942   5554.   5406.   7821.  10233   2812.     0
## 2 arrows   50    54.2    61     60.3    67     70     6.47     0
## 3 targets  15     35     43     42.7    50.8    64    10.5     0
## 4 acc      0.3    0.604   0.743   0.708    0.830   0.98    0.155     0
## # i 1 more variable: hist <named list>
```

The mean for the number of days of experience is 5406. The mean for the accuracy is 0.708

4.10 Standardising Numerical Variables

```
archery_preprocess <- preProcess(archery)
archery_preprocess
```

```
## Created from 450 samples and 6 variables
##
## Pre-processing:
## - centered (4)
## - ignored (2)
## - scaled (4)
```

```
archery_stardize <- predict(archery_preprocess, archery)
archery_stardize
```

```
## # A tibble: 450 x 6
##   id      commenced      xp  arrows targets  acc
##   <ord> <date>      <dbl>   <dbl>   <dbl> <dbl>
## 1 2      2019-09-25 -1.31 -0.816 -1.30 -1.17
## 2 140    2019-01-16 -1.22 -0.0433 -0.923 -1.02
## 3 185    1997-01-14  1.64  0.729  1.74  1.49
## 4 469    2012-11-25 -0.421 -0.507 -0.543 -0.383
## 5 153    2017-08-08 -1.03  0.111 -0.638 -0.763
## 6 172    2003-08-12  0.785  0.266  0.978  0.946
## 7 412    2004-12-31  0.605 -0.507  0.503  0.864
## 8 517    1997-03-09  1.62  0.266  1.26  1.26
## 9 424    2003-04-19  0.826  0.884  1.45  1.10
## 10 184    2018-09-20 -1.18 -0.816 -1.21 -1.05
## # i 440 more rows
```

```
dim(archery_stardize)
```

```
## [1] 450 6
```

4.11 Comparing Standardised Variables

```
inspect_num(archery_stardize)
```

```
## # A tibble: 4 x 10
##   col_name   min      q1 median      mean      q3    max    sd pcnt_na hist
##   <chr>    <dbl>  <dbl>  <dbl>    <dbl> <dbl> <dbl> <dbl>  <dbl> <named list>
## 1 xp      -1.75 -0.876 0.0529  3.07e-17 0.859  1.72    1      0 <tibble>
## 2 arrows -1.59 -0.931 0.111  -1.85e-16 1.04   1.50    1      0 <tibble>
## 3 targets -2.63 -0.733 0.0275  2.38e-16 0.764  2.02    1      0 <tibble>
## 4 acc     -2.64 -0.676 0.221   1.83e-16 0.789  1.75    1      0 <tibble>
```

5 Variable with higher mean.

The 2 variables have means very close to zero. We can say that their means are approximately zero.

6 Variable with higher median.

accuracy has higher median of 0.221

7 Variable with higher inter-quartile range

```
IQR(archery_stardize$xp)
```

```
## [1] 1.735277
```

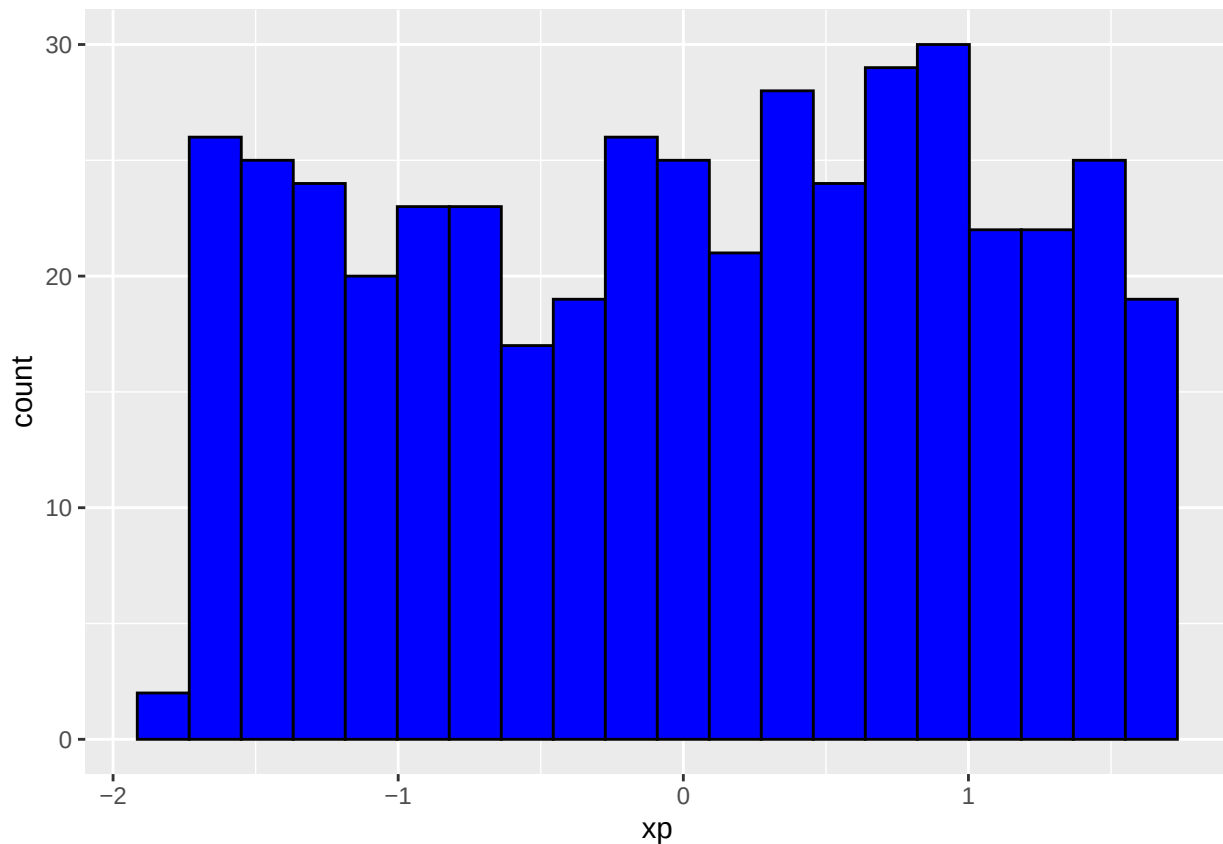
```
IQR(archery_stardize$acc)
```

```
## [1] 1.464058
```

Experience has higher inter-quartile range of 1.74

8 Plot histograms of each variable and calculate the skewness.

```
ggplot(archery_stardize, aes(xp))+  
  geom_histogram(bins=20,fill="blue",col="black")
```

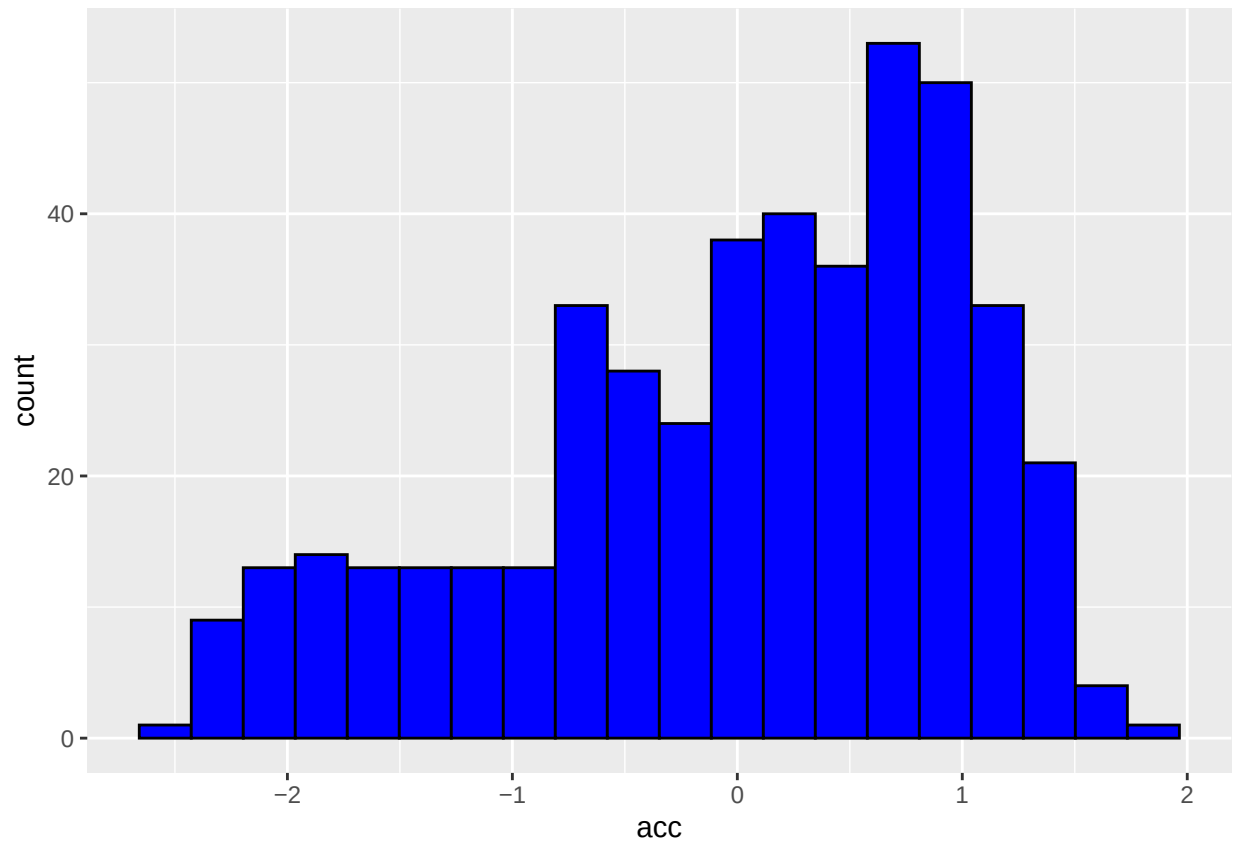


```
moments::skewness(archery_stardize$xp)
```

```
## [1] -0.08051181
```

The shape of histogram shows similarity to uniform distribution with some fluctuation of the bars. This might be due to the club recruits relative similar number of members in over time.

```
ggplot(archery_stardize, aes(acc))+  
  geom_histogram(bins=20,fill="blue",col="black")
```



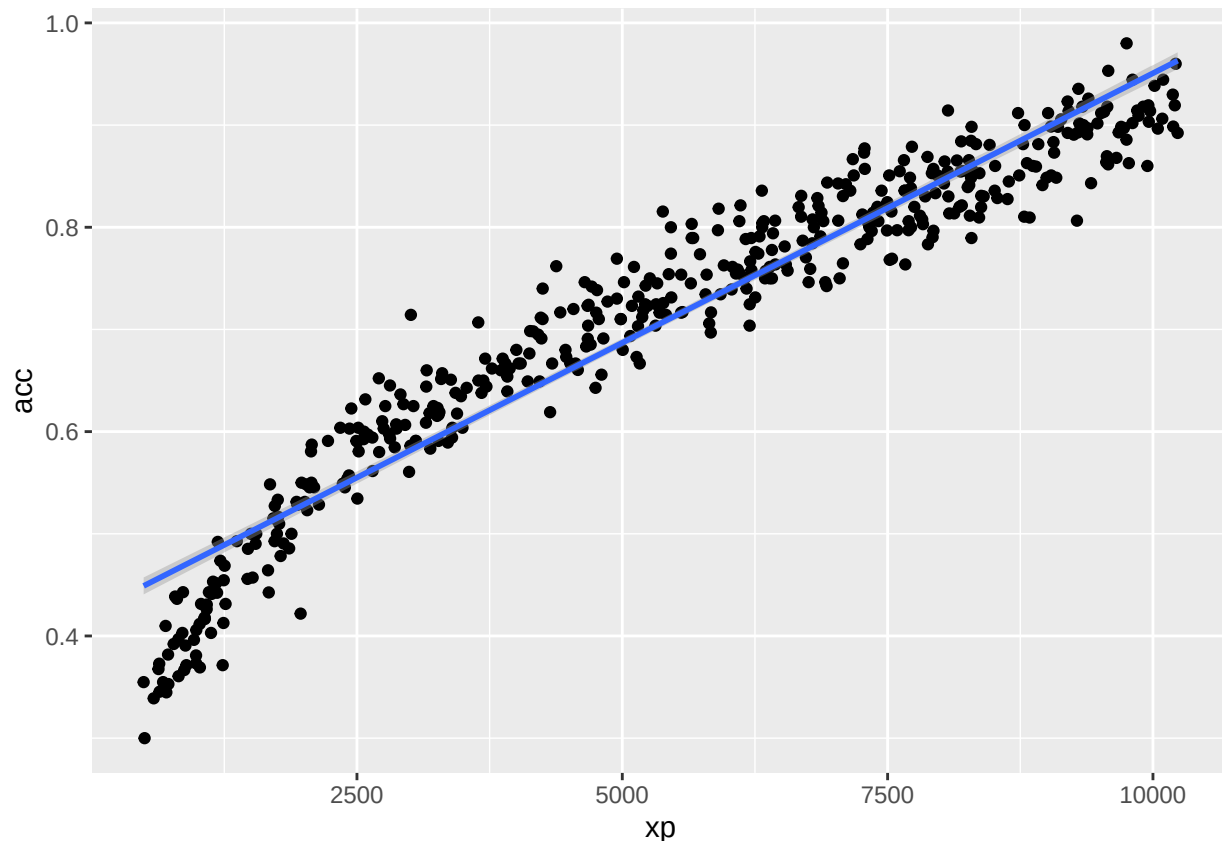
```
moments::skewness(archery_stardize$acc)
```

```
## [1] -0.635975
```

The histogram of accuracy seems to indicate left-skewed distribution. This indicates higher tendency of higher accuracy.

8.1 Experience and Accuracy Relationship

```
ggplot(archery, aes(x=exp, y=acc))+  
  geom_point()+  
  geom_smooth(method="lm")
```



There is a positive relation between experience and accuracy. However, there seems to be a curve in the relationship. When the archer just started archery and had less experience, the smooth line over estimate the accuracy of archery. The relationship makes sense as we expect accuracy to increase with days of experience (more practices). For new starter, it might be difficult to hit the target. However, as an archer gains more and more experience, it might be difficult to improve accuracy, resulting in the curve as shown. From the curve in the plot, It coheres with what we found in Q9 as accuracy is left-skewed. The slope of the curve is steep and become more gentle with increase in experience.

8.2 Box-Cox Transformation

```
archery_bc <- BoxCoxTrans(y = archery$acc, x = archery$xp, lambda=seq(-10,10,0.1))
archery_bc
```

```
## Box-Cox Transformation
##
## 450 data points used to estimate Lambda
##
## Input data summary:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.3000  0.6038  0.7426  0.7084  0.8304  0.9800
##
```

```
## Largest/Smallest: 3.27
## Sample Skewness: -0.634
##
## Estimated Lambda: 2.4
```

```
archery_bc$lambda
```

```
## [1] 2.4
```

8.3 Applying the Transformation

```
archery <- mutate(archery, acc_bc = predict(archery_bc, archery$acc))
archery
```

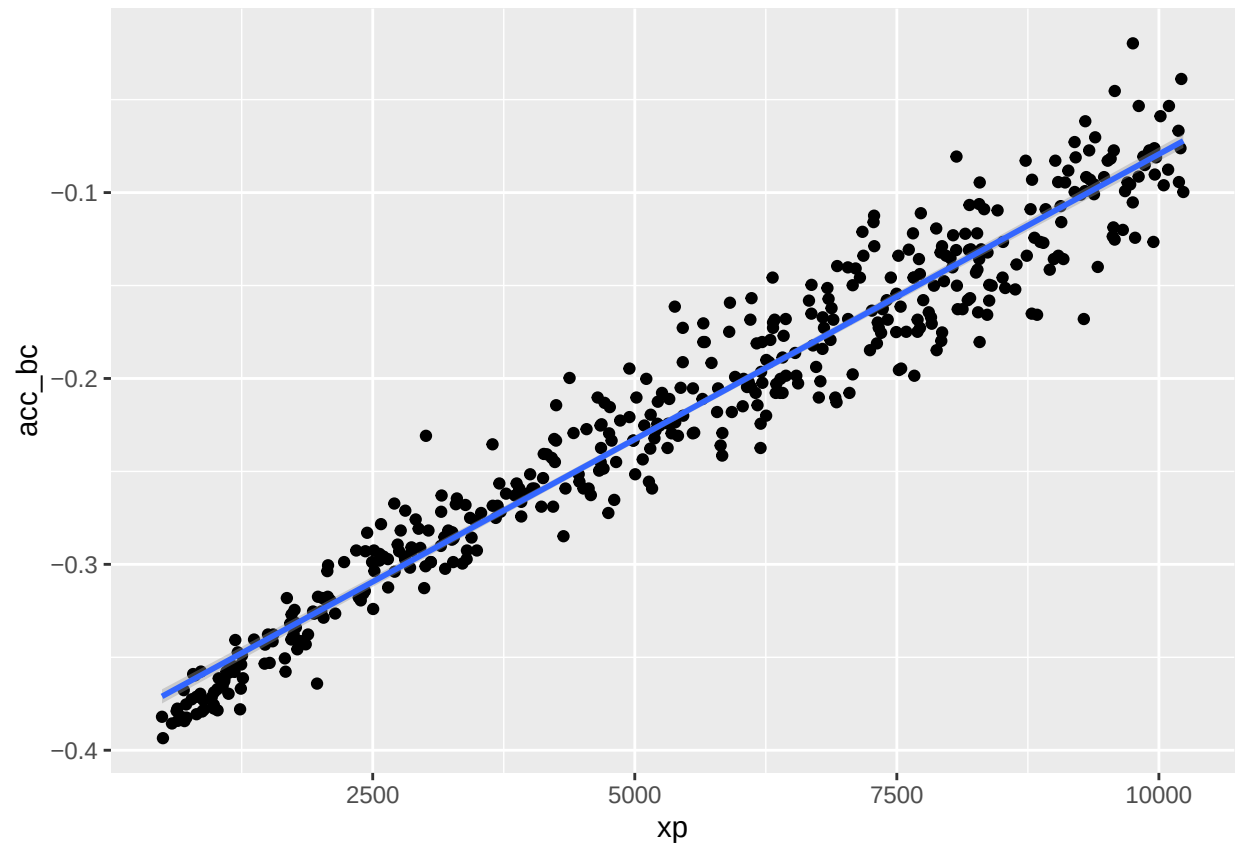
```
## # A tibble: 450 x 7
##   id      commenced      xp arrows targets  acc  acc_bc
##   <ord> <date>      <int> <int>   <int> <dbl> <dbl>
## 1 2      2019-09-25  1726    55     29 0.527 -0.327
## 2 140    2019-01-16  1978    60     33 0.55  -0.317
## 3 185    1997-01-14 10015    65     61 0.938 -0.0589
## 4 469    2012-11-25  4221    57     37 0.649 -0.269
## 5 153    2017-08-08  2504    61     36 0.590 -0.299
## 6 172    2003-08-12  7614    62     53 0.855 -0.131
## 7 412    2004-12-31  7107    57     48 0.842 -0.141
## 8 517    1997-03-09  9961    62     56 0.903 -0.0903
## 9 424    2003-04-19  7729    66     58 0.879 -0.111
## 10 184   2018-09-20  2096    55     30 0.545 -0.319
## # i 440 more rows
```

```
dim(archery)
```

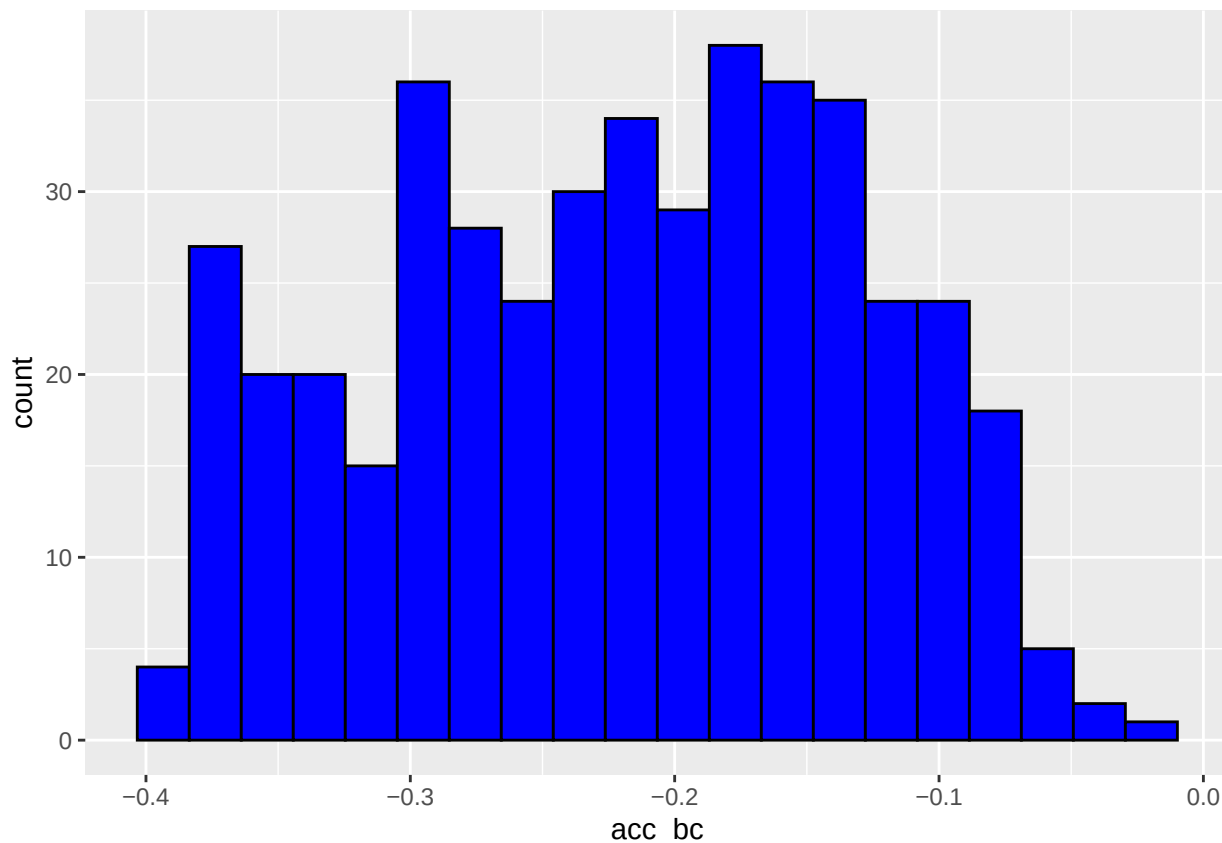
```
## [1] 450 7
```

8.4 Transformed Data

```
ggplot(archery, aes(x=xp, y=acc_bc))+
  geom_point()+
  geom_smooth(method="lm")
```



```
ggplot(archery, aes(acc_bc))+  
  geom_histogram(bins=20,fill="blue",col="black")
```



```
moments :: skewness(archery$acc_bc)
```

```
## [1] -0.1267762
```

The plot shows a better linear fit of the data, even though, it still slightly overestimate the accuracy for archery with less experience. The histogram still appears to have slight left skewness. However, the skewness is reduced. The value of skewness for accuracy shift from -0.636 in standardized data to -0.127 (closer to zero) when Box-Cox transformation is applied to the data.

8.5 Population Linear Model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

y_i is the accuracy

x_i is the number of days of experience

β_0 is the intercept. When experience is 0, accuracy is estimated to be β_0

β_1 is the slope. when number of day of experience change by 1, accuracy is estimated to change by β_1

ϵ_i is an error term which is the difference between the actual accuracy and the estimated accuracy from the model.

8.6 Sample Regression Line

$$y_i^* = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

y_i^* is the Box-cox transformed accuracy

x_i is the number of days of experience

$\hat{\beta}_0$ is the intercept. When experience is 0, accuracy is estimated to be $\hat{\beta}_0$

$\hat{\beta}_1$ is the slope. when number of day of experience change by 1, accuracy is estimated to change by $\hat{\beta}_1$

8.7 Fitting the Linear Model

```
archery_lm <- lm(acc_bc ~ xp, data = archery)
summary(archery_lm)
```

```
##
## Call:
## lm(formula = acc_bc ~ xp, data = archery)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.066715 -0.013075 -0.000494  0.012608  0.067233
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.861e-01  2.087e-03 -184.99  <2e-16 ***
## xp           3.067e-05  3.426e-07   89.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02041 on 448 degrees of freedom
## Multiple R-squared:  0.9471, Adjusted R-squared:  0.947
## F-statistic: 8016 on 1 and 448 DF,  p-value: < 2.2e-16
```

$$y^* = -0.386 + 3.07 \times 10^{-5}x$$

y^* is the Box-cox transformed accuracy

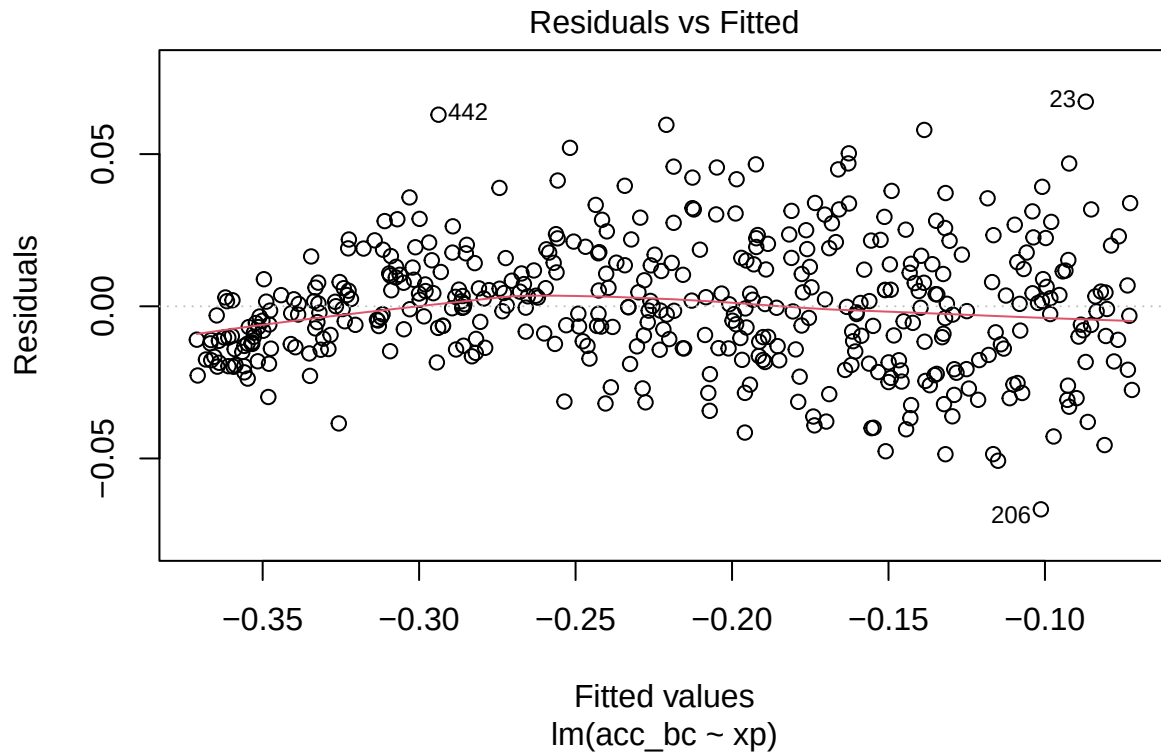
x is the number of days of experience

-0.386 is the intercept. When experience is 0, the Box-cox transformed accuracy is estimated to be -0.386

3.07×10^{-5} is the slope. when number of day of experience change by 1, the Box-cox transformed accuracy is estimated to change by 3.07×10^{-5} # Model Diagnostics

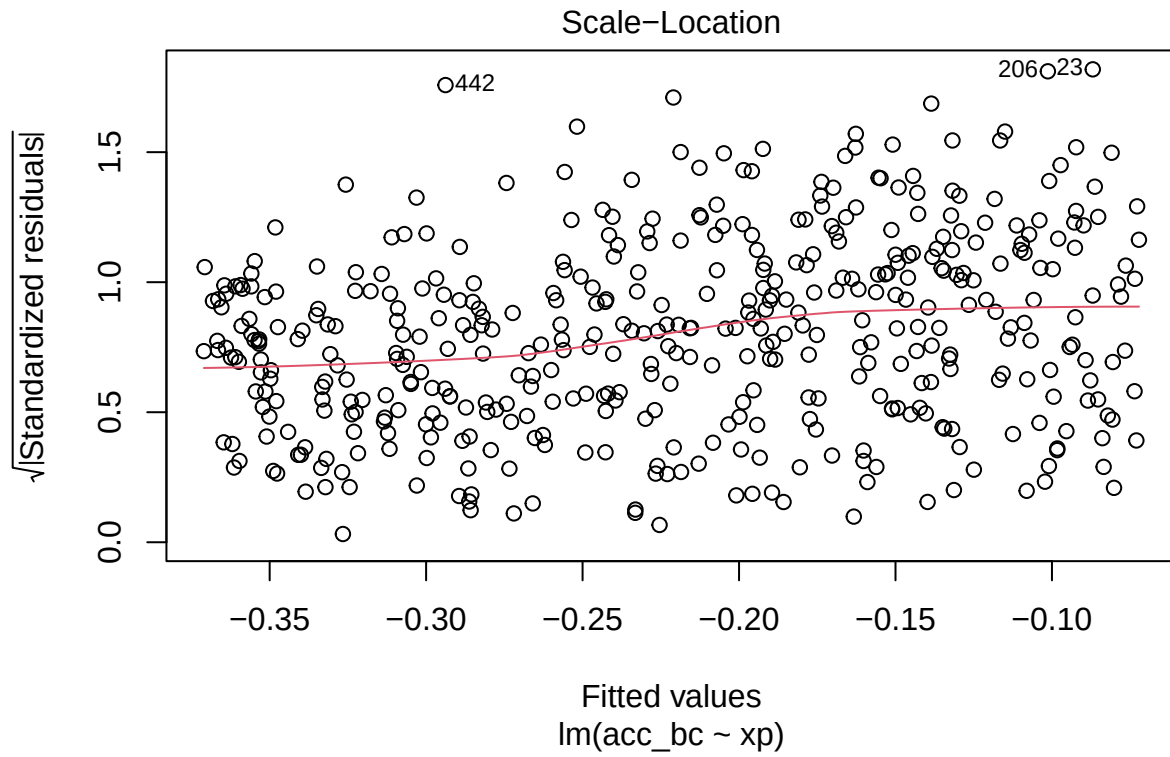
The fitted linear model is assessed against the four assumptions discussed in the course: linearity, constant variance, normally distributed errors, and independence. These checks are used to identify limitations rather than to discard the model required for the assignment.

```
plot(archery_lm, which=1) #plot to check residual versus fitted plot, check for linearity
```



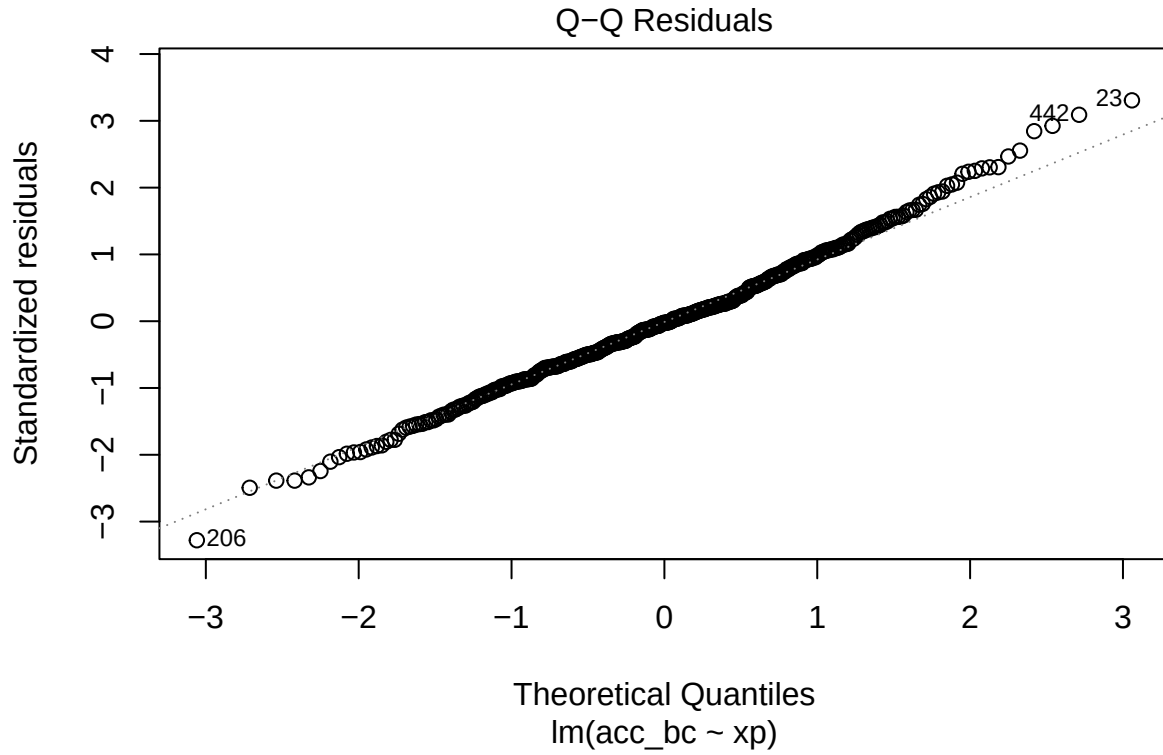
There is no obvious trend in the scatter plot. The line is mostly straight. It might be affected a small number of outliers. This satisfies the assumption of linearity.

```
plot(archery_lm, which=3) # check Homoscedasticity, noise has the same variance
```



There is no apparent trend in the scatter plot. The red line appear to be straight. It indicates that the model has constant spread. Therefore, it satisfies the assumption of constant variance or homoscedasticity.

```
plot(archery_lm, which=2) # check if the noise normally distributed
```



From the plot, the points lie along the dotted line with only tails slightly drifted from the dotted line. Hence, it satisfies the assumption that the noise terms are normally distributed.

The fourth assumption is the error terms are independent. From the data that we collected, we only have experience as predictor. However, there could be other factors that affect the accuracy such as gender or physical fitness which are included into to the error terms making the error terms to correlate with the response variable.

9 Predictions experience of 2, 5, 10, 15, 20 and 25 years.

```

xp_new <- tibble(xp = c(2*365.25,5*365.25,10*365.25,15*365.25,20*365.25,25*365.25))
#assume that there are 356*4+1 days in 4 years, so 1 year, there are 365.25 days
accuracy_new <- predict(archery_lm, xp_new)
accuracy_new <- round((accuracy_new*archery_bc$lambda+1)^(1/archery_bc$lambda),3)
#convert to original scale
accuracy_new

```

```

##      1      2      3      4      5      6
## 0.424 0.520 0.640 0.734 0.815 0.885

```

```

confi <- predict(archery_lm, xp_new, interval = "confidence", level = 0.9)
confi_convert <- round((confi*archery_bc$lambda+1)^(1/archery_bc$lambda),3)
#convert to original scale
confi_convert <-tibble(fit= confi_convert[, "fit"],
                      lwr=confi_convert[, "lwr"], upr=confi_convert[, "upr"] )
#create a tibble
confi_convert <- mutate(confi_convert, "diff_upr_lwr"= upr-lwr,
                      years_of_xp=c(2,5,10,15,20,25))
confi_convert <- relocate(confi_convert, "years_of_xp", .before=fit)
confi_convert <- mutate(confi_convert, years_of_xp=as.integer(years_of_xp))
#change to integer for years of experience
confi_convert

```

```

## # A tibble: 6 x 5
##   years_of_xp  fit   lwr   upr diff_upr_lwr
##         <int> <dbl> <dbl> <dbl>         <dbl>
## 1             2 0.424 0.413 0.434         0.0210
## 2             5 0.52  0.513 0.526         0.0130
## 3            10 0.64  0.636 0.643         0.00700
## 4            15 0.734 0.732 0.737         0.00500
## 5            20 0.815 0.812 0.817         0.00500
## 6            25 0.885 0.882 0.888         0.00600

```

The accuracy for an archer with experience of 2 years is 0.424. We are 90% confident that an average archer with experience of 2 years will have accuracy between 0.413 and 0.434

The accuracy for an archer with experience of 5 years is 0.520. We are 90% confident that an average archer with experience of 5 years will have accuracy between 0.513 and 0.526

The accuracy for an archer with experience of 10 years is 0.640. We are 90% confident that an average archer with experience of 10 years will have accuracy between 0.636 and 0.643

The accuracy for an archer with experience of 15 years is 0.734. We are 90% confident that an average archer with experience of 15 years will have accuracy between 0.732 and 0.737

The accuracy for an archer with experience of 20 years is 0.815. We are 90% confident that an average archer with experience of 20 years will have accuracy between 0.812 and 0.817

The accuracy for an archer with experience of 25 years is 0.885. We are 90% confident that an average archer with experience of 25 years will have accuracy between 0.882 and 0.888

10 Discussion and Conclusion

10.1 Interpretation

```
inspect_num(archery)
```

```
## # A tibble: 5 x 10
##   col_name      min      q1    median     mean      q3      max      sd pcnt_na
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 xp          490     2942    5554.    5406.    7821.    1.02e+4  2.81e+3      0
## 2 arrows       50       54.2     61      60.3     67       7 e+1  6.47e+0      0
## 3 targets      15       35      43      42.7     50.8     6.4 e+1  1.05e+1      0
## 4 acc          0.3      0.604    0.743    0.708    0.830     9.8 e-1  1.55e-1      0
## 5 acc_bc     -0.393   -0.293   -0.213   -0.220   -0.150   -1.97e-2  8.86e-2      0
## # i 1 more variable: hist <named list>
```

A good archer needs to have accuracy more than mean accuracy 0.708 and It also depends on how we define what a good archer is. If a good archer is a person who has accuracy more or equal to 0.885, it would take 25 years to be a good archery. We also observe that the confidence interval is wider for archers with 2 and 5 years of experience as compared to archers with experience of 10 years and above. This suggests that our prediction of accuracy for archers with fewer years of experience tend be less accurate than those with more experience.

10.2 Limitations

There is also limitation for this linear model. It suggests that as number of years of experience increases, the accuracy will further increase. However, this might not be the case in reality as the archer's health could affect the accuracy as the archer grow older. Hence, this model might only be accurate for a certain range of number of years of experience. Further, we observe that when number of years of experience increases, the accuracy will exceed 1 which is wrong as accuracy is ratio of targets and arrows, so it cannot exceed 1. Below is the example of calculation for 40 years of experience to substantiate the claim.

```
xp40 <- tibble(xp = 40*365.25)
accuracy40 <- predict(archery_lm, xp40)
accuracy40 <- round((accuracy40*2.4+1)^(1/2.4),3) #lambda=2.4, convert to original scale
accuracy40
```

```
##      1
## 1.06
```

The accuracy for 40 years of experience is 1.06 which is more than 1.

10.3 Conclusion

The analysis provides evidence of a strong positive association between archery experience and shooting accuracy. In the sample, predicted accuracy increased from 42.4% at two years of experience to 88.5% at 25 years.

However, these predictions should not be interpreted as a definitive training timeline. The concept of a “good archer” requires a clearly defined performance threshold, and the model includes experience as its only explanatory variable. In addition, the linear model eventually produces impossible accuracy values above 1 when extrapolated too far beyond the observed data.

Overall, the results suggest that **experience is an important predictor of accuracy, with the largest improvements occurring earlier in an archer’s development**, but becoming a good archer depends on more than experience alone.

10.4 Appendix: Reproducible R Analysis

All R code used to produce the analysis is retained in the report. This preserves the original coursework workflow while presenting the results in a structure suitable for a professional data analysis report.